Harshit Kashyap Sarma

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Profile

Audio machine learning and DSP researcher pursuing an MSc in Sound and Music Computing at UPF's Music Technology Group **(MTG)**. Experience in real-time audio plugins (C++/JUCE), psychoacoustic feature modeling (Essentia), and neural audio models (PyTorch). Visharad-certified tabla player with 10+ years of training in expressive microtiming and percussive acoustics.

Education

MSc in Sound and Music Computing, Universitat Pompeu Fabra (MTG), Barcelona *Sep 2026 – Present*
Coursework: Digital Signal Processing for Sound and Music, Machine Learning for Sound and Music, Music Perception and Cognition

BSc in Computer Science (AI/ML specialization), VIT, Vellore, India CGPA: 8.37/10 *Jul 2023 – May 2026*

Visharad in Tabla (classical percussion) *Completed 2019*

Research Experience

Music Technology Group, UPF – Audio DSP Research Contributor *Jul 2026 – Present*
Supervisor: Dr. Frederic Font

- Ported the psychoacoustic sharpness descriptor to Essentia, reducing benchmark error from **24.21% to 0.36%** with a **1.41**× speedup over the original SciPy implementation.
- Implemented the ISO 532B (DIN 45631, Zwicker) specific-loudness pipeline, with Butterworth and Chebyshev filter coefficients precomputed for 28 third-octave bands to ease translation to C++.
- Fixed cascade-deletion race conditions and post-delete signal handlers in the Freesound platform (Django/PostgreSQL), and wrote recovery scripts to repair affected database state.

Audio & Machine Learning Projects

Real-Time Source Separation of Tabla (Dayan & Bayan) | *PyTorch, Librosa, causal neural networks* *Ongoing*

- Developing causal, low-latency neural models to separate the pitched dayan from the low-frequency, transient-heavy bayan into distinct stems.
- Evaluating separation quality (SDR) against processing latency, targeting live use without transient smearing or phase cancellation.

Generative Kick Drum Synthesizer | *Python, PyTorch, Librosa* [GitHub]

- Trained a VAE on 128×128 Mel spectrograms to encode percussive attacks into a continuous latent space, enabling smooth morphing between clicky and resonant kick textures.

Oscillator Synthesizer (VST3) | *C++, JUCE* [GitHub]

- Built a real-time multi-waveform synth with anti-aliased oscillators, lock-free parameter queues and sample-accurate MIDI handling across DAW buffer blocks.
- Wrote *SynthQuest*, a published 7-part tutorial series on DSP foundations, envelope generators and thread safety.

Engineering Experience

KnowledgeVerse AI – Software Development Engineer Intern *May 2025 – Aug 2025; Dec 2025 – Present*

- Migrated backend services from synchronous to asynchronous (`asyncio`/`aioboto3`), removing event-loop blocking in high-throughput reporting pipelines.
- Built a Redis caching layer with custom TTL invalidation, cutting document retrieval latency by ∼**15**×.
- Published the official Python client SDK (`kv-platform`) to PyPI with typed, versioned releases.

Technical Skills

Audio ML & DSP: PyTorch, torchaudio, Librosa, Essentia, SciPy, NumPy, VAEs, spectral analysis, IIR/FIR filter design
Audio Systems: C/C++, JUCE (VST3), Python, ChucK, real-time audio buffers
Infrastructure: Linux, Git, Docker, Redis, FastAPI, AWS (Lambda, S3), RESTful APIs

Leadership

Board Member & R&D Lead, CodeChef-VIT *2024 – 2026*

- Ran workshops on digital audio basics, filter design and JUCE plugin development for 100+ students.
- Led technical infrastructure for flagship hackathons (DevSoc, CookOff), sustaining 99.9% uptime for 1,000+ participants.